

Dhruv Patel

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EDUCATION

University of California, Los Angeles (UCLA)

Jun. 2027

B.S. Mathematics of Computation, Data Science Engineering Minor

GPA: 3.77/4.00

Coursework: Machine Learning, Deep Learning & Neural Networks, ML for Finance (Graduate), AI for Scientific Computing (Graduate), NLP, Real Analysis, Probability, Linear Algebra, Data Structures, Algorithms, Numerical Methods, Theory of Computing

Honors: Upsilon Pi Epsilon (CS Honor Society), UCLA Dean's Honors List (3×), Eagle Scout

Organizations: Association for Computing Machinery, UCLA Poker Group, UCLA Cubing Club, Intramural Basketball

EXPERIENCE

UCLA Mathematical & Statistical Finance Group

Los Angeles, CA

Quantitative Research Assistant

Jun. 2025 – Present

- Researching ML and AI applications in quantitative finance under Prof. Mihai Cucuringu
- Forecasted daily volume dispersion over a ~4k stock universe (2000–2024); HAR-based model cut MSE by 28.5% vs AR(1) baseline
- Devised a market-neutral long-short equity strategy using volume dispersion forecasts; 1.5+ Sharpe over 8-year OOS backtest
- Designed a configurable strategy backtesting pipeline, reaching 4× speedup via vectorized NumPy operations and 5D PnL tensor

Pacific Life

Newport Beach, CA

Data Science Intern

Jun. 2026 – Sep. 2026

- Implemented a Cortex Search Service to filter medical records down to only the information relevant to underwriting guidelines
- Flagged discrepancies between filtered records and applicant answers via LLM, projected to cut underwriter caseload by ~10%
- Built an LLM agent that tunes this workflow's prompt, model type, and similarity threshold to optimize false omission rate
- Deployed an Azure DevOps pipeline pulling per-user Snowflake spend, sending AWS SNS alerts when daily spend exceeds \$50

Smart Circle International

Newport Beach, CA

AI Engineer Intern

Sep. 2025 – Jun. 2026

- Developed a Snowflake Cortex LLM agent for D2D sales territory reporting and zip code-level lead replacement recommendations
- Engineered a Python pipeline to parse and structure SCORM e-learning content across 10+ formats for AI training workflows
- Aggregated 15+ campaign documents across formats and structures into a centralized knowledge base for LLM ingestion

Analytics Intern

Jun. 2025 – Sep. 2025

- Trained Python churn prediction models (Random Forest & Logistic Regression), achieved 50%+ recall with 85%+ accuracy
- Synthesized 100K+ dispositions into a weekly Tableau report on sales efficiency by office; adopted across 15+ sales campaigns
- Defined office-level metrics evaluating AI-driven recruiting; informed funding allocation decisions across 100+ offices

Bruin Sports Analytics

Los Angeles, CA

Tennis Data & Scouting Analyst

Oct. 2024 – Jun. 2026

- Built a transformer model to predict shot type and court location from sequential rally data using PyTorch
- Produced 20+ visualizations of serving tendencies used by the UCLA D1 Men's Tennis Team to adjust pre-match strategy
- Standardized end-to-end workflow from match footage to court visualizations, enabling adoption across the scouting team

PROJECTS

LOBSTER News Alpha Agent | *Python (pandas, SciPy, NumPy), Anthropic API*

- Created an LLM agent combining news sentiment with limit-order-book flow for intraday signals across 2,879 events
- Beat lexicon baseline on inference-heavy TSLA headlines (IC +0.28, $p < 0.001$), the only subgroup surviving BH-FDR

NBA Timeout Optimization via Hidden Markov Models | *Python (hmmlearn, NumPy, pandas, Scikit-Learn)*

- Fit a 3-state HMM on 250K+ NBA possessions to infer latent momentum regimes from sequential scoring data
- Found timeout impact peaks in Q4 when trailing 6-10 (+0.14 points/possession) across 12K+ within-team matched pairs

EEG Motor Imagery Classification | *Python (PyTorch, NumPy, SciPy)*

- Benchmarked CNN, LSTM, GRU, TCN, and Transformer on 22-channel EEG under 36-fold leave-one-subject-out validation
- Showed compact models (4K parameters) generalize better than 200K-parameter alternatives across 9 subjects

SKILLS

Languages: Python, C++, Java, C, SQL, R, JavaScript, HTML/CSS, LaTeX

Libraries: NumPy, pandas, Scikit-Learn, PyTorch, SciPy, hmmlearn, Optuna, LightGBM, XGBoost, Matplotlib, Seaborn

Tools: Snowflake, Azure DevOps, AWS, Docker, Git, Linux, JupyterLab, Tableau, Excel, Google Cloud